
Stack (continued)

CSE-207

Infix Notation

- To add A, B, we write

$$A+B$$

- To multiply A, B, we write

$$A*B$$

- The operators ('+' and '*') go in between the operands ('A' and 'B')
- This is "*Infix*" notation.

Prefix Notation

- Instead of saying "A plus B", we could say "add A,B " and write
 $+ A B$
- "Multiply A,B" would be written
 $* A B$
- This is *Prefix* notation.

Postfix Notation

- Another alternative is to put the operators after the operands as in

$A B +$

and

$A B *$

- This is *Postfix* notation.

-
- The terms infix, prefix, and postfix tell us whether the operators go between, before, or after the operands.



Pre A In B Post

- Infix expression is called **polish notation**
- Postfix expression is called **reverse polish notation**

Parentheses

- Evaluate $2+3*5$.

- + First:

$$(2+3)*5 = 5*5 = 25$$

- * First:

$$2+(3*5) = 2+15 = 17$$

- Infix notation requires Parentheses.

What about Prefix Notation?

- $+ 2 * 3 5 =$
 $= + 2 \underline{* 3 5}$
 $= \underline{+ 2 15} = 17$
- $* + 2 3 5 =$
 $= * \underline{+ 2 3 5}$
 $= \underline{* 5 5} = 25$
- No parentheses needed!

Postfix Notation

- $2\ 3\ 5\ *\ + =$
 $= 2\ \underline{3\ 5\ *} +$
 $= \underline{2\ 15\ +} = 17$
- $2\ 3 + 5\ * =$
 $= \underline{2\ 3 +}\ 5\ *$
 $= \underline{5\ 5\ *} = 25$
- No parentheses needed here either!

-
- Infix is the only notation that requires parentheses in order to change the order in which the operations are done.

Fully Parenthesized Expression

- A FPE has exactly one set of Parentheses enclosing each operator and its operands.
- Which is fully parenthesized?

$(A + B) * C$

$((A + B) * C)$

$((A + B) * (C))$

Infix to Prefix Conversion

Move each operator to the left of its operands & remove the parentheses:

$$((A + B) * (C + D))$$


Infix to Prefix Conversion

Move each operator to the left of its operands & remove the parentheses:

(+ A B * (C + D))



Infix to Prefix Conversion

Move each operator to the left of its operands & remove the parentheses:

* + A B (C + D)



Infix to Prefix Conversion

Move each operator to the left of its operands & remove the parentheses:

* + A B + C D

Order of operands does not change!

Infix to Postfix

$(((A + B) * C) - ((D + E) / F))$



$A \ B + \ C * \ D \ E + \ F / -$

- Operand order does not change!
- Operators are in order of evaluation!

Infix to Postfix

Algorithm: Polish (Q, P)

Suppose Q is an arithmetic expression written in infix notation. This algorithm finds the equivalent postfix expression P.

1. Push “(“ onto STACK, and add “)” to the end of Q.
2. Scan Q from left to right and repeat step 3 to 6 for each element of Q until the STACK is empty.
3. If an operand is encountered, add it to P.
4. If a left parenthesis is encountered, push it onto STACK.
5. If an operator is encountered, then:
 - a) Repeatedly POP from STACK and add to P each operator (on the top of STACK) which has the same precedence as or higher precedence than that operator.
 - b) Add that operator to STACK.[End of if structure]
6. If a right parenthesis is encountered, then:
 - a) Repeatedly pop from the STACK and add to P each operator until a left parenthesis is encountered.
 - b). Remove the left parenthesis.[Do not add the left parenthesis to P].[End of if structure]
[End of step 2 loop].
7. EXIT.

FPE Infix to Postfix

$(((A + B) * (C - E)) / (F + G))$



- stack: <empty>
- output: []

FPE Infix to Postfix

(((A + B) * (C - E)) / (F + G))



Symbol	stack	Output
	(	[]

FPE Infix to Postfix

$((A + B) * (C - E)) / (F + G))$



Symbol	stack	Output
(	((	

FPE Infix to Postfix

$(A + B) * (C - E) / (F + G)$



Symbol	stack	Output
(	((	
(	((	

FPE Infix to Postfix

$A + B) * (C - E)) / (F + G))$



Symbol	stack	Output
(	((	
(	((((	
(	(((((	

FPE Infix to Postfix

+ B) * (C - E)) / (F + G))



Symbol	stack	Output
(	((	
(	((	
(	((	
A	((	A

FPE Infix to Postfix

B) * (C - E)) / (F + G))



Symbol	stack	Output
(	((	
(	(((	
(	((((	
A	((((	A
+	((((+	A

FPE Infix to Postfix

) * (C - E)) / (F + G))



Symbol	stack	Output
(	((	
(	((	
(	((	
A	((	A
+	((	A
B	((	AB

FPE Infix to Postfix

) * (C - E)) / (F + G))



Symbol	stack	Output
B	(((((+	AB

FPE Infix to Postfix

 * (C - E)) / (F + G))

Symbol	stack	Output
B	((((+	AB
)	((((	AB +

FPE Infix to Postfix

(C - E)) / (F + G))



Symbol	stack	Output
B	((((+	AB
)	((()	AB+
*	(((*	AB+

FPE Infix to Postfix

C - E)) / (F + G))



Symbol	stack	Output
B	((((+	AB
)	((()	AB+
*	(((*	AB+
(	(((*	AB+

FPE Infix to Postfix

- E)) / (F + G))



Symbol	stack	Output
B	(((((+	AB
)	((((	AB+
*	((((*	AB+
(	((((*	AB+
C	(((* (	AB+C

FPE Infix to Postfix

E)) / (F + G))



Symbol	stack	Output
B	(((((+	AB
)	((((	AB+
*	(((*	AB+
(	(((*	AB+
C	(((*	AB+C
-	(((*(-	AB+C

FPE Infix to Postfix

E)) / (F + G))



Symbol	stack	Output
-	(((*(-	AB+C

FPE Infix to Postfix

)) / (F + G)))



Symbol	stack	Output
-	(((*(-	AB+C
E	(((*(-	AB+CE

FPE Infix to Postfix

) / (F + G)))



Symbol	stack	Output
-	(((*(-	AB+C
E	(((*(-	AB+CE
)	(((*	AB+CE -

FPE Infix to Postfix

/ (F + G))



Symbol	stack	Output
-	(((*(-	AB+C
E	(((*(-	AB+CE
)	(((*	AB+CE-
)	((	AB+CE- *

FPE Infix to Postfix

(F + G))



Symbol	stack	Output
-	(((*(-	AB+C
E	(((*(-	AB+CE
)	(((*	AB+CE-
)	((	AB+CE-*
/	((/	AB+CE-*

FPE Infix to Postfix

F + G))



Symbol	stack	Output
-	(((*(-	AB+C
E	(((*(-	AB+CE
)	(((*	AB+CE-
)	((	AB+CE-*
/	((/	AB+CE-*
(	((/(	AB+CE-*

FPE Infix to Postfix

F + G))



Symbol	stack	Output
(	((/(	AB+CE-*

FPE Infix to Postfix

+ G))



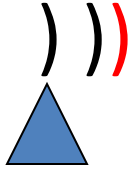
Symbol	stack	Output
(	((/(	AB+CE-*
F	((/(	AB+CE-*F

FPE Infix to Postfix

G))

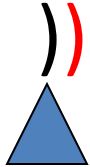

Symbol	stack	Output
(	((/(	AB+CE-*
F	((/(	AB+CE-*F
+	((/(+	AB+CE-*F

FPE Infix to Postfix



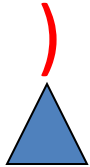
Symbol	stack	Output
(	((/(	AB+CE-*
F	((/(	AB+CE-*F
+	((/(+	AB+CE-*F
G	((/(+	AB+CE-*FG

FPE Infix to Postfix



Symbol	stack	Output
(	((/(	AB+CE-*
F	((/(	AB+CE-*F
+	((/(+	AB+CE-*F
G	((/(+	AB+CE-*FG
)	((/	AB+CE-*FG +

FPE Infix to Postfix



Symbol	stack	Output
(	((/(	AB+CE-*
F	((/(	AB+CE-*F
+	((/(+	AB+CE-*F
G	((/(+	AB+CE-*FG
)	((/	AB+CE-*FG+
)	((	AB+CE-*FG+ /

FPE Infix to Postfix



Symbol	stack	Output
)	(	AB+CE-*FG+/

FPE Infix to Postfix



Symbol	stack	Output
)	(	AB+CE-*FG+/
)		AB+CE-*FG+/

Example: Q: $4 * (5 + 3) - 24 / 6$ and P: ?

Infix Expression Q	Stack	Postfix Expression P
4	(	4
*	(*	4
(	(* (	4
5	(* (	4, 5
+	(* (+	4, 5
3	(* (+	4, 5, 3
)	(*	4, 5, 3, +
-	(-	4, 5, 3, +, *
24	(-	4, 5, 3, +, *, 24
/	(- /	4, 5, 3, +, *, 24
6	(- /	4, 5, 3, +, *, 24, 6
)	(-)	4, 5, 3, +, *, 24, 6, /
		4, 5, 3, +, *, 24, 6, /, -

Postfix Expression P : 4, 5, 3, +, *, 24, 6, /, -



END